思考题1：

$$D_{n+1}=\begin{vmatrix} x & 1 & & & & \\ n & x & 2 & & & \\ & n-1 & x & \cdots & & \\ & & \cdots & \cdots & n-1 & \\ & & & 2 & x & n \\ & & & & 1 & x \end{vmatrix}$$

记原行列式为$\mathrm{D}_{n+1}(x)$

将每一行都加上后面的所有行，再提取第一行公因子

$$\mathrm{D}_{n+1}(x)=(x+n)\begin{vmatrix} 1 & 1 & 1 & \cdots & 1 & 1 \\ n & x+n-1 & x+n & \cdots & x+n & x+n \\ 0 & n-1 & x+n-2 & \cdots & x+n & x+n \\ \cdots & \cdots & \cdots & \cdots & \cdots & \cdots \\ 0 & 0 & 0 & \cdots & x+1 & x+n \\ 0 & 0 & 0 & \cdots & 1 & x \end{vmatrix}$$

从最后一列到第二列，每一列都减去前面一列

$$\mathrm{D}_{n+1}(x)=(x+n)\begin{vmatrix} 1 & 0 & 0 & \cdots & 0 & 0 \\ n & x-1 & 1 & \cdots & 0 & 0 \\ 0 & n-1 & x-1 & \cdots & 0 & 0 \\ \cdots & \cdots & \cdots & \cdots & \cdots & \cdots \\ 0 & 0 & 0 & \cdots & x-1 & n-1 \\ 0 & 0 & 0 & \cdots & 1 & x-1 \end{vmatrix}$$

$=(x+n)D_n(x-1)$　　(双递推，不仅n在变化，x也在变化)

$=(x+n)\,(x+n-2)\,(x+n-2*2)\cdots(x+n-2(n-1))D_1(x-n)$

$=\dfrac{(x+n)!!}{(x-n)!!}$

思考题 2：

$$\mathrm{D}_n=\begin{vmatrix} \frac{1}{a_1+b_1} & \frac{1}{a_1+b_2} & \cdots & \frac{1}{a_1+b_n} \\ \frac{1}{a_2+b_1} & \frac{1}{a_2+b_2} & \cdots & \frac{1}{a_2+b_n} \\ \cdots & \cdots & \cdots & \cdots \\ \frac{1}{a_n+b_1} & \frac{1}{a_n+b_2} & \cdots & \frac{1}{a_n+b_n} \end{vmatrix}$$

把前 $n-1$ 列每列减去第 n 列

$$D_n=\begin{vmatrix} \frac{b_n-b_1}{(a_1+b_1)(a_1+b_n)} & \frac{b_n-b_2}{(a_1+b_2)(a_1+b_n)} & \cdots & \frac{1}{a_1+b_n} \\ \frac{b_n-b_1}{(a_2+b_1)(a_2+b_n)} & \frac{b_n-b_2}{(a_2+b_2)(a_2+b_n)} & \cdots & \frac{1}{a_2+b_n} \\ \cdots & \cdots & \cdots & \cdots \\ \frac{b_n-b_1}{(a_n+b_1)(a_n+b_n)} & \frac{b_n-b_2}{(a_n+b_2)(a_n+b_n)} & \cdots & \frac{1}{a_n+b_n} \end{vmatrix}$$

$$=\frac{\prod_{i=1}^{n-1}(b_n-b_i)}{\prod_{j=1}^{n-1}(a_j+b_n)}\begin{vmatrix} \frac{1}{a_1+b_1} & \frac{1}{a_1+b_2} & \cdots & 1 \\ \frac{1}{a_2+b_1} & \frac{1}{a_2+b_2} & \cdots & 1 \\ \cdots & \cdots & \cdots & \cdots \\ \frac{1}{a_n+b_1} & \frac{1}{a_n+b_2} & \cdots & 1 \end{vmatrix}$$

再将前 $n-1$ 行减去最后一行

$$D_n=\frac{\prod_{i=1}^{n-1}(b_n-b_i)}{\prod_{j=1}^{n-1}(a_j+b_n)}\begin{vmatrix} \frac{b_n-b_1}{(a_1+b_1)(a_1+b_n)} & \frac{b_n-b_2}{(a_1+b_2)(a_1+b_n)} & \cdots & 0 \\ \frac{b_n-b_1}{(a_2+b_1)(a_2+b_n)} & \frac{b_n-b_2}{(a_2+b_2)(a_2+b_n)} & \cdots & 0 \\ \cdots & \cdots & \cdots & \cdots \\ \frac{1}{a_n+b_1} & \frac{1}{a_n+b_2} & \cdots & 1 \end{vmatrix}$$

$$=\frac{\prod_{i=1}^{n-1}(b_n-b_i)(a_n-a_i)}{\prod_{j=1}^{n-1}(a_j+b_n)\prod_{k=1}^{n-1}(a_n+b_k)}D_{n-1}$$

$$=\frac{\prod_{1\leqslant i<j\leqslant n}(a_j-a_i)(b_j-b_i)}{\prod_{i,j=1}^{n}(a_i+b_j)}$$

思考题 3：已知 $\begin{vmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \cdots & \cdots & \cdots & \cdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{vmatrix} = 1$，$a_{ij} = -a_{ji}$，

求行列式 $\begin{vmatrix} a_{11}+b & a_{12}+b & \cdots & a_{1n}+b \\ a_{21}+b & a_{22}+b & \cdots & a_{2n}+b \\ \cdots & \cdots & \cdots & \cdots \\ a_{n1}+b & a_{n2}+b & \cdots & a_{nn}+b \end{vmatrix}$

记原式为 D_n(行列式等于 1 的那个)

$D_n = (-1)^n D_n$

故 n 为奇数时 $D_n = 0$，原式中 n 为偶数

所求式 $= \frac{1}{b} \begin{vmatrix} b & b & b & \cdots & b \\ 0 & a_{11}+b & a_{12}+b & \cdots & a_{1n}+b \\ 0 & a_{21}+b & a_{22}+b & \cdots & a_{2n}+b \\ \cdots & \cdots & \cdots & \cdots & \cdots \\ 0 & a_{n1}+b & a_{n2}+b & \cdots & a_{nn}+b \end{vmatrix}$

第二行到 $n-1$ 行都减去第一行

所求式 $= \frac{1}{b} \begin{vmatrix} 0+b & b & b & \cdots & b \\ -b & a_{11} & a_{12} & \cdots & a_{1n} \\ -b & a_{21} & a_{22} & \cdots & a_{2n} \\ \cdots & \cdots & \cdots & \cdots & \cdots \\ -b & a_{n1} & a_{n2} & \cdots & a_{nn} \end{vmatrix} = 0 + 1 = 1$

思考题 4：

$$D_n=\begin{vmatrix} \frac{1-a_1^n b_1^n}{1-a_1 b_1} & \frac{1-a_1^n b_2^n}{1-a_1 b_2} & \cdots & \frac{1-a_1^n b_n^n}{1-a_1 b_n} \\ \frac{1-a_2^n b_1^n}{1-a_2 b_1} & \frac{1-a_2^n b_2^n}{1-a_2 b_2} & \cdots & \frac{1-a_2^n b_n^n}{1-a_2 b_n} \\ \cdots & \cdots & \cdots & \cdots \\ \frac{1-a_n^n b_1^n}{1-a_n b_1} & \frac{1-a_n^n b_2^n}{1-a_n b_2} & \cdots & \frac{1-a_n^n b_n^n}{1-a_n b_n} \end{vmatrix}$$

$$D_n=\begin{vmatrix} 1 & a_1 & a_1^2 & \cdots & a_1^{n-1} \\ 1 & a_2 & a_2^2 & \cdots & a_2^{n-1} \\ 1 & a_3 & a_3^2 & \cdots & a_3^{n-1} \\ \cdots & \cdots & \cdots & \cdots & \cdots \\ 1 & a_n & a_n^2 & \cdots & a_n^{n-1} \end{vmatrix} * \begin{vmatrix} 1 & b_1 & b_1^2 & \cdots & b_1^{n-1} \\ 1 & b_2 & b_2^2 & \cdots & b_2^{n-1} \\ 1 & b_3 & b_3^2 & \cdots & b_3^{n-1} \\ \cdots & \cdots & \cdots & \cdots & \cdots \\ 1 & b_n & b_n^2 & \cdots & b_n^{n-1} \end{vmatrix}$$

$$=\prod_{1\leqslant j<i\leqslant n}(a_i-a_j)\,(b_i-b_j)$$